

Xujiang Tang

Theoretical Mathematics Researcher; Machine Learning Theory Researcher
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RESEARCH INTERESTS

Theoretical mathematics centered on algebraic topology, finite group theory, equivariant homotopy theory, fixed point methods, formal proof and Lean, and long-range structures related to the Langlands program. Secondary research direction: foundational machine learning theory, including generalization, learnability, out-of-distribution prediction, optimization, stochastic processes, and dynamical systems.

COLLABORATORS AND MENTORS

- William Redman, Johns Hopkins University - <https://engineering.jhu.edu/faculty/william-redman/>
- Ziyue Qiao, Great Bay University - <https://qiaoziyue.com/>

EDUCATION

Yangtze University College of Arts and Sciences, Jingzhou, China

- B.Sc. in Mathematics and Applied Mathematics, expected June 2027. Selected coursework: mathematical analysis, linear algebra, probability and statistics, differential equations, computational modeling, optimization theory, numerical computation, data structures, Python, R, and MATLAB.
- Selected marks: Optimization Theory 97/100; Numerical Computation Methods 96/100.

VISITING POSITION AND TRAINING

Visiting Student, Great Bay University, 2025-present

- Research in theoretical mathematics and machine learning theory.

Zhejiang University Lean Formalization Summer School, 2026

- Formal proof, Lean, Mathlib, and computer-assisted theorem proving.

RESEARCH EXPERIENCE

Finite Group Theory and Quillen's p-Subgroup Conjecture, 2025-present

- Work on centralizer energy, FRUL-WOS spheres, subgroup complexes, and p-local group-theoretic structures related to Quillen's p-subgroup conjecture.

Algebraic Topology and Equivariant Homotopy Theory, 2025-present

- Work on geometric fixed point tomography, spoke Bokstedt periodicity, equivariant fixed point methods, and homotopical reconstruction from local or fixed point data.

Formal Proof and Lean, 2026-present

- Study of Lean formalization, proof engineering, and the long-term formalization of advanced mathematical arguments in algebra, topology, and analysis.

Theoretical Machine Learning and Optimization, 2024-present

- Work on learnability, generalization guarantees, out-of-distribution prediction, and optimization-theoretic aspects of learning algorithms.

Stochastic Processes, Dynamical Systems, and Quantitative Finance, 2023-present

- Research on stochastic-process models, rough volatility, MCMC-based simulation, Koopman/operator-theoretic dynamics, and robust financial decision problems.

PUBLICATIONS AND MANUSCRIPTS

Submitted manuscripts

- X. Tang. Centralizer Energy, FRUL-WOS Spheres, and Quillen's p-Subgroup Conjecture. Submitted to Annals of Mathematics.
- X. Tang. Geometric Fixed Point Tomography and Spoke Bokstedt Periodicity. Submitted to Algebraic & Geometric Topology.
- X. Tang and C. Fan. Out-of-Distribution Generalization Error Bounds for Successful Prediction of Deep Autoregressive Algorithms. Submitted to NeurIPS 2026.
- X. Tang. Quartic Difficulty: Assessing the Learnability of Unsupervised Learning Algorithms. Submitted to COLT.

Published and accepted papers

- Q. Li and X. Tang. HKGEduRec: A Knowledge Graph-Enhanced Dynamic Hybrid Framework for Educational Recommendation with Cold-Start Mitigation. IEEE ICMEIM 2025. EI-indexed.
- X. Tang and Q. Li. Logical Gene Encoding: A Bio-Inspired Approach for Energy-Efficient Automated Reasoning. IEIT 2025. EI-indexed. DOI: 10.2991/978-94-6463-803-5_81.
- Q. Li and X. Tang. Robust Optimal Reinsurance and Investment with Inflation Risk: A Game-Theoretic Approach and Explicit Solutions. AIMS Mathematics. Accepted, 2025.

HONORS, COMPETITIONS, AND ACTIVITIES

- Kaggle Silver Medal, Jigsaw Toxic Comment Classification Challenge.
- Regional Gold Medal, WorldQuant International Quant Championship.
- Third Prize, UCAS Graduate AI Forum.
- Mathematical Modeling Competition, Hua Zhong Cup.
- National College Students Statistical Modeling Competition.
- Multiple national second prizes in mathematical modeling competitions.
- Reviewer, ICLR 2026 Workshop.

PROFESSIONAL EXPERIENCE

Peking University - Open-Source LLM Group, Jun.-Aug. 2024

- Contributed to LLM deployment and retrieval-augmented generation systems.

Deloitte, Jul.-Sep. 2024

- Professional experience in consulting workflows, teamwork, and communication.

Guolian Securities, Jul.-Sep. 2024

- Applied statistical methods to securities research and quantitative workflows.

SKILLS

- Mathematics: algebraic topology, finite group theory, homotopy theory, representation theory, optimization, numerical analysis, stochastic processes, differential equations, causal inference, statistical modeling.
- Formal proof and programming: Lean, Mathlib, Python, PyTorch, NumPy, Pandas, MATLAB, R, C++, LaTeX, Git/GitHub, Linux.
- Languages: Mandarin Chinese, English.